QuickSyn SCPI Controller and Logger — version 0.0.5

Date: 2026-10-11 · **Target:** Python 3.12, Qt.py → PySide6 · **Status:** verified against documentation, a built-in simulator and real Veusz 3.6.2 / 4.2.1 (and patched 4.2.1) on Python 3.12.3 and 3.13.16; **not yet run against hardware**

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Licence	GPL-3.0-or-later (SPDX), full text in `LICENSE`; no associated project
Versioning	Semantic Versioning 2.0.0 [15]; supersedes 0.0.4
Archive	`quicksyn_scpi_controller_v0.0.5.7z` (13 files; SHA-256 `93d1c95b70d9125d1bab9e0305ff9723226964decacd9241bdac015a83ff0697`)
Reusable module	`claude/vszh5_writer.py` 0.0.5 in this project (SHA-256 `2cdf43d2…ae33c1c`); export script `claude/quicksyn_veusz_export_v0_0_5.py`
Companion release	`isp_speed_monitor_v1.7.2.7z` (script 1.3.1, `render_in_veusz.sh` 1.0.2; SHA-256 `959be0a2494d89d5a834ca18cf53ab7aef40a8a146f3fb66fb400435215c6ba9`) — commercial AdaptEd product, fixed directly, no GPL code imported

What changed in 0.0.5

Item	Detail
Live SCPI console	View ▸ SCPI console (Ctrl+Shift+C), docked under the tabs: `HH:MM:SS.mmm`, source `GUI` / `POLL`, `→` command, `←` reply with latency, `✕` error; colours legible in light and dark themes; telemetry polls hidden by default; filter, pause/resume, auto-scroll, copy, save, clear. Queued signal from the sampler thread, 150 ms batched rendering, caps 5000 lines / 20 000 exchanges (≈ 5 MB measured). `transactions.csv` stays the full record
Render: Windows paths	PNG path bound on its own line outside the `exec()` string (0.0.4 unescaped it twice: `C:\Users…` failed, `\n` became a newline, a quote broke the string)
Render: fallback	`method="auto"` falls back to the Veusz Python package when the program **fails**, not only when missing; both errors reported if both fail
Render: robustness	A program that exits before reading its script (broken pipe, a race) or cannot be started → `RuntimeError`, not `BrokenPipeError` / `OSError`; stdout pipe closed; Veusz never left running after an error
Upstream Veusz patch	`upstream/veusz-4.2.1-vszh5-load-listen-quit.patch` + `upstream/VEUSZ_UPSTREAM_ISSUE.md` in the archive (see below)
Docs	README §5.4 (console), §6.2/§6.5 (fixes, patch), §10 rows, §11 history; licence open item closed; history entries ascending in every file

Findings from 0.0.4 — status

#	Finding	Status
1	`veusz --export file.vszh5` and embedded `Load()` cannot open `.vszh5` (`CommandInterpreter.Load()` reads every file as UTF-8 text)	**Fixed upstream-style**: patch makes `Load()` detect the HDF5 signature (offset 0, 512, 1024, …) and call `Document.load(mode='hdf5')`. Patched 4.2.1: PNG and 2-page PDF from the command line, embedded Load+Export OK, text `.vsz` unchanged; unpatched: segfault (139) / `IndexError`. Same code still on `master` 2026-10-11 [22]. Not yet submitted
2	Veusz 4.2.1 does not exit on `Quit()` in `--listen --quiet` (`quitProgram()` only closes a never-shown window)	**Fixed upstream-style**: patch adds `QCoreApplication.instance().quit()`; exits in 0.46 s (unpatched: still running after 30 s). Packages keep the marker + terminate workaround for released Veusz
3	`isp_speed_monitor.py` 1.3.0: render menu used invalid `--export … --export-dpi … %PAGE%`; `render_in_veusz.sh` relied on `timeout 180`; `vz_text` turned `\` into a line break, left `[ ]`	**Fixed in ISP 1.3.1 / bundle 1.7.2**: `render_vszh5_pages()` (listen + `loadHDF5Doc`, marker, paths outside `exec`), `--render-veusz FILE [--outdir]`, `vz_text` = Veusz `latexEscape`, script 1.0.2 uses an end marker. Also fixed a docstring `\b` (backspace) escape in its own history text
4	Ubuntu `/usr/bin/veusz` starts `/usr/bin/python3`; fails (`No module named 'PyQt5.sip'`) where that is not the packaged Python	**Handled**: QuickSyn `auto` falls back in-process; ISP reports Veusz's own error + `veusz --version` / env-var hint

Scope and hard constraints

Full SCPI control of the NI / Phase Matrix QuickSyn family over the USB virtual COM port, plus a timestamped log of every command generated and a telemetry stream at a user-set sample rate. All of [1, Table 5] is implemented.

- **115200 8N1, no flow control, CR (0x0D) terminator, 64-byte device buffer including the terminator** — fixed on USB. `DIAG:BAUD` retargets the debug port, **not** USB.
- **SCPI needs firmware ⩾ 100**; SPI is native-only.
- Only `*IDN?`, `*RST`, `*SAV`, `*RCL` of IEEE 488.2 exist — **no error queue, no `*OPC` / `*ESR?`**; acceptance is confirmed by read-back.

- Mandatory delays applied automatically: 2 ms reset, 100 ms `*SAV`, 50 ms `*RCL`, 200 ms `LIST:ERAS`, 300 ms flash list point, 100 µs RAM point, 50 ms + 2.5 ms/pt `LIST:SAV`, seconds for the reference DAC, 1 ms `FREQ` with FM on.
- Lists and sweeps refused while FM is on; dwell quantised to 5 µs; power to 0.1 dB.

Millihertz overflows the buffer in compound commands (69 B for a 20 GHz `SWE:FAST:FREQ:SETUP`). `format_frequency()` picks the shortest exact spelling among mlHz/KHz/MHz/GHz; `_check_compound_length()` refuses anything that still will not fit.

Veusz output (unchanged since 0.0.4)

`.vszh5` written directly with h5py by `vszh5_writer.py` (author's method from `isp_speed_monitor.py` 1.3.0 [21]); no text `.vsz` (7 significant digits: $q = 10^{k-6}$ GHz, $|\varepsilon| \leq q/2 \rightarrow \leqslant 500$ Hz below 10 GHz, $\leqslant 5$ kHz above). Tagged `raw/`, `events/`, `stability/`, `meta/`; overview + 6 pages; commanded values overlaid. The controller writes it at stop, on demand (Ctrl+E) and every N min; separate process. Render via `--listen` + `loadHDF5Doc`, Python-package fallback.

Documentation conflicts and resolutions

Conflict	Resolution
Post-reset blanking `ON` [1] vs `off` [2]	Never assumed; `STAT?` read after every reset
SCPI `*IDN?` field order undefined	All five raw fields shown; firmware gate advisory
Query double-issue unstated for SCPI	`TEMPERATURE_DOUBLE_QUERY = True`, recorded
Response terminator unspecified	Reader accepts CR / LF / CRLF
0.01 dB [4] vs tenth-dBm [1]	Quantised to 0.1 dB
Missing comma in `SWE:FAST:FREQ:SETUP` example	Eight fields + optional `RUN`
"Start NF Sweep" typo	Treated as `...FREQ:STAR 1`
32767 vs "32,000" list points	32767 for range checks
`DIAG:BAUD` vs USB	Debug port only
mlHz default vs GHz examples	Shortest exact spelling, pre-flighted

Verification (0.0.5)

Check	Result
`run_all.py` (release 102/102, smoke incl. console and GUI `.vszh5` export, reduction, `check_vszh5`)	PASS on 3.12.3 and 3.13.16, also from the extracted archive
`check_vszh5.py` with real Veusz	41/41 with 4.2.1, patched 4.2.1 and 3.6.2 (new: Windows-style path with quote via stub `Export`, broken program `program` / `auto`, cannot-start, render into `odd \dir 'q'`)
New path test vs 0.0.4 writer	fails on 0.0.4 (`SyntaxError`) — confirms the regression test
SCPI console	93–106 exchanges, 76–89 polls hidden by default, filter, pause/resume, ✕ error line, save; screenshots light/dark inspected
ISP 1.3.1	`-W error` compile clean; `--selftest` OK; `gui_smoke.py` 9 shots; `--render-veusz` 7 PNGs with 4.2.1; quick-exit, non-executable and missing Veusz → clear error, exit 1; `render_in_veusz.sh` 7 PNGs with 4.2.1 (3.3 s), patched 4.2.1 (1.3 s), 3.6.2 (1.3 s), broken exe exit 1
Patch	`patch -p1 --dry-run` clean on 4.2.1 sdist
Archives	7-Zip 16.02 `t` OK; extraction identical to sources; ISP file list identical to 1.7.1

Unverified: Windows render (does the windowed `veusz.exe` read a piped script in `--listen` mode? fallback covers it); a real `mamba env create`; anything needing the instrument.

Open items

1. Copyright holder / NRAO release approval (files name William W. Wallace).
2. NRAO/GBO header template, if one is mandated.
3. First hardware session: capture `transactions.csv` (and the console text) and the `*IDN?` line.
4. Windows: create env, log a short session, Veusz ▸ *Write .vszh5 and render*; try `--render-method program` to see whether the program path itself works.
5. File the Veusz issue + patch at [22] (needs a GitHub account).

6. ISP (not changed): Download panel legend overlaps data.

Closed: licence of the `isp_speed_monitor.py`-derived code — the author owns Adaptive Electrodynamics LLC dba AdaptEd; README §13 updated in 0.0.5.

References

#	Reference	URL
[1]	National Instruments, "QuickSyn Microwave Synthesizers: Communications Specifications," Doc. No. 5580510-01, 2014.	Project upload
[2]	National Instruments, "QuickSyn Microwave Frequency Synthesizer: User Guide," Doc. No. 5580513-01, Rev. F.	Project upload
[3]	National Instruments, "QuickSyn … Calibration Software User's Manual," Doc. No. 5580515-01, Rev. B, 2014.	Project upload
[4]	National Instruments, "QuickSyn Frequency Synthesizers," brochure and datasheet, Oct. 2017.	Project upload
[5]	National Instruments, "Clearing QuickSyn Memory," application note.	Project upload
[6]	IEEE Std 488.2-1992, *IEEE Standard Codes, Formats, Protocols, and Common Commands…*, IEEE, 1992.	https://ieeexplore.ieee.org/document/8705775
[7]	IEC 60488-2:2004 / IEEE 488.2, *Standard digital interface for programmable instrumentation — Part 2*, IEC, 2004.	https://www.vde-verlag.de/iec-normen/preview-pdf/info_iec60488-2%7Bed1.0%7Den.pdf
[8]	SCPI Consortium, *SCPI Volume 1: Syntax and Style*, Version 1999.0, May 1999.	https://www.ivifoundation.org/downloads/SCPI/scpi-99.pdf
[9]	IEEE Std 1139-2008, *IEEE Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology*, IEEE, 2008.	https://ieeexplore.ieee.org/document/4797525
[10]	IVI Foundation, "Specification Downloads."	https://www.ivifoundation.org/specifications/default.html

#	Reference	URL
[11]	Veusz Developers, "Veusz — a scientific plotting package."	https://veusz.github.io/
[12]	Phase Matrix, Inc., 20 GHz QuickSyn press release, IEEE MTT-S IMS 2010, May 2010.	https://www.rfcafe.com/miscellany/press-releases/2010/phase-matrix-5-20-2010.htm
[13]	Free Software Foundation, "GNU General Public License, version 3," 29 Jun. 2007.	https://www.gnu.org/licenses/gpl-3.0.html
[14]	SPDX Workgroup, "GPL-3.0-or-later," SPDX License List.	https://spdx.org/licenses/GPL-3.0-or-later.html
[15]	T. Preston-Werner, "Semantic Versioning 2.0.0."	https://semver.org/spec/v2.0.0.html
[16]	Python Software Foundation, "What's New In Python 3.13."	https://docs.python.org/3/whatsnew/3.13.html
[17]	Anaconda, Inc., conda-forge pages for `qt.py`, `veusz`, `h5py`.	https://anaconda.org/conda-forge/veusz
[18]	Veusz Developers, "Veusz 4.2.1" source, PyPI.	https://pypi.org/project/veusz/4.2.1/
[19]	A. Collette *et al.*, "h5py 3.16.0," PyPI.	https://pypi.org/project/h5py/
[20]	`7zip-bin` npm package (p7zip 16.02).	https://www.npmjs.com/package/7zip-bin
[21]	W. W. Wallace, *isp_speed_monitor.py* 1.3.0, Adaptive Electrodynamics LLC dba AdaptEd, Oct. 2026.	Supplied by the author (not public)
[22]	Veusz Developers, "veusz/veusz," GitHub repository (`master`: `document/commandinterpreter.py`, `veusz_listen.py`) and issue tracker, accessed 11 Oct. 2026.	https://github.com/veusz/veusz ; https://github.com/veusz/veusz/issues